

Scenario-Robust Traffic State Reconstruction and Queue Forecasting

IEEE BigData Cup 2026 — TrafficFlowBench · Team Steins · final submission V13 (public leaderboard 0.88107)

1. Summary

Validation (March) and private (April) are **new traffic scenarios**: the scenario seed is redrawn, so per-link free-flow speeds, queued-speed levels, ramp demand and the bottlenecks that activate all change. A model that is validated only on held-out training days cannot see this shift. Our solution is built around one discipline: **every change is accepted only on out-of-scenario evidence**, measured on the published validation/private layers. We use them for evaluation only, never for fitting a prediction. With that discipline in place, three structural findings carried most of the gain:

1. **Measurement noise has a corridor-wide common mode.** The free-flow branch of the generator's fundamental diagram is flat, so a free-flowing speed minus its link plateau is pure measurement noise. That noise shares a component across the whole corridor at each 5-minute slot, and the other links observed at the same slot estimate it.
2. **Scenario statistics must come from the scenario itself.** Plateaus and profiles measured on the training months plant a per-link bias in the new months. Measured on the split's own published layer (Task 1 is offline, so this is allowed), they remove about 0.3 km/h of speed RMSE out of scenario.
3. **Published layers carry sensors that were overlooked.** The masked layer publishes the Task 2 origin row T at about 58% coverage. And on-ramp flow collapses while the mainline link it feeds is queued, while the ramp layer stays published inside the 90-minute mainline blackouts that follow each Task 2 origin.

From the inherited pipeline (public 0.87446) the final submission reaches **0.88107**. Our out-of-scenario evaluators predicted the last two online deltas to within 0.0005.

2. Tasks, rules and validation

Score. $S = 0.35 \cdot S_{\text{state}} + 0.30 \cdot S_{\text{queue}} + 0.15 \cdot S_{\text{physics}} + 0.20 \cdot S_{\text{ODME}}$. Task 3 is computed from the Task 1 answer.

Information sets. We follow the organizers' ruling (forum topic 742068). Task 2 is online: a forecast may only use released data with a timestamp at or before the origin T, in any file. Task 1 is offline: any released observation of the split may be used. Our Task 2 inputs are the 60-minute window history, the masked-layer origin row at T, and the same-day and earlier-day masked layer before T. Ramp values inside the horizon and mainline observations after the post-horizon buffer are never read for Task 2.

Out-of-scenario evaluators.

- *Task 1, hidden cells.* We hide 3% of the observed eligible cells of validation/private (about 39k per panel-split) and rebuild every feature without them. This includes the split-own plateaus and profiles, so the hidden values never enter their own predictors. We then score the predictions against the published values (tfb/t1_shift_eval.py). Synthetic blackouts (all links blank for 18 slots) score the gap specialist the same way.
- *Task 2, mined events.* We take onset events and ongoing windows found in the validation/private masked layer and build their inputs exactly as at inference (tfb/t2_onset_shift.py, tfb/t2_prod_shift.py). Ongoing windows use the organizers' selector rule (persistence IoU ≤ 0.9).

Adoption rule. A paired gain of more than 2 standard errors. Both splits must be non-negative. At least half of the changed panels must improve in both splits. The gain must survive dropping the best panel.

Change	Predicted Δ	Online Δ	Evaluator
V10a $\rightarrow$ V12	+0.004 ... +0.0055	-0.0012	train-day holdout (blind to the scenario shift)
V12 $\rightarrow$ V12b	+0.006	+0.0058	out-of-scenario hidden cells
V12b $\rightarrow$ V13	+0.0016	+0.0021	out-of-scenario hidden cells

The V12 miss is the reason for the protocol. Its Task 1 changes were right in the training scenario and wrong in the new ones (Section 4.2). The out-of-scenario evaluator caught this, and its predictions now match the online deltas.

3. What the generator looks like

- **Two flat plateaus.** In free flow, $v/v_f = 1.00$ at every flow level (p10-p90 within $\pm 1.5\%$). A queue sits at about $0.30\text{--}0.35 \cdot v_f$. There is no speed precursor of breakdown.
- **Queues appear as blocks.** Whole clusters of links switch on at once (for example links 4–10 of D7_I10_E). Before onset, the flow/capacity of the cluster that will queue is ordinary (0.5–0.7). The highest-flow links never queue.
- **Common-mode noise.** Free-flow speed noise has sd 0.8–1.75 km/h per panel and is nearly white in time (lag-1 autocorrelation 0.06). It is correlated about 0.3 between links 1 and 30 positions apart. The common factor has sd 0.46–1.09 km/h. Flow noise has no usable common mode.
- **Ramp sensor.** On-ramp inflow falls to 5–70% of its time-of-day profile while the mainline link it feeds is queued, and off-ramp flow rises about 10%.
- **Scenario shift.** Compared with train, per-link plateaus in validation/private move with sd about 1.3 km/h. Per-link queued-speed levels move with sd 1–6 km/h. Bottleneck clusters switch on or off; for example, D12_I5_S links 43–45 queue on 89% of train days but 45% of March days, and the onset time of D12_I5_N moves from 12.4 h to 10.8 h.

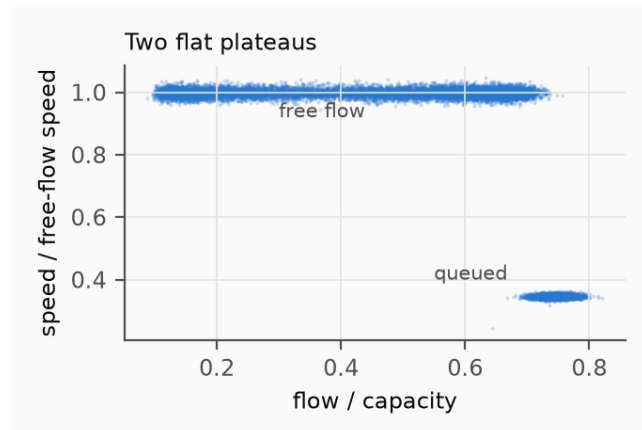


Figure 1. Speed against flow for one link of D7_I10_E (train, 15k cells): a flat free-flow branch and a flat queued plateau, no breakdown precursor.

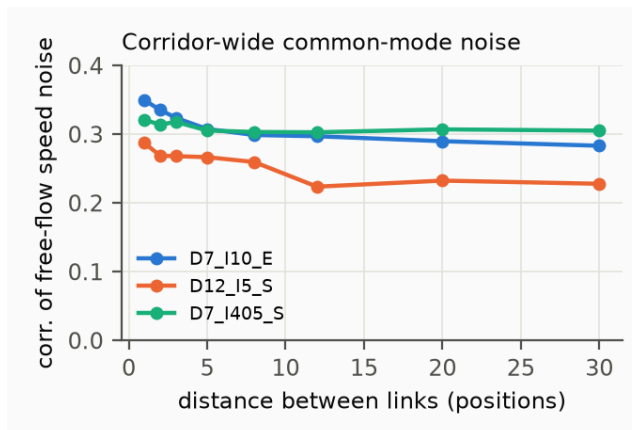


Figure 2. Correlation of free-flow speed noise between links k positions apart (train). It stays near 0.3 up to 30 positions: a corridor-wide common mode.

4. Task 1 — state reconstruction

4.1 Model

A LightGBM residual model is fitted over temporal linear interpolation: speed in km/h, and flow per lane. Features: temporal/spatial interpolations, $\pm 1/\pm 2$ neighbours, neighbour-difference estimates, time-of-day profiles, local level relative to the profile, pct_observed and link attributes. On top of these:

- **Common mode.** cf_sp is the mean of $(v - plateau)$ over the free-flowing links observed at the same slot. We also use a relative version, a ± 5 -link local version and the plateau itself. Only the speed model uses them; they did not help flow.
- **Split-own statistics.** Plateaus and the gap specialist's ramp and speed/flow profiles are measured on the split's own published masked layer.
- **Gap specialist.** Validation/private blank every link for $T+5 \dots T+90$ after each Task 2 origin. Train has no such blackouts, so we place synthetic ones at onset and ongoing origins in train. A dedicated model is trained on the cells inside them, with ramp-ratio features: per-link on/off-ramp flow divided by its profile, temporal smoothing, $\pm 2/\pm 5$ -link windows, and the share of collapsed on-ramps nearby.
- **Capacity.** Out of scenario, the learning curve kept improving up to all train target cells (about 20M rows, with the matrix filled column-wise into one float32 array).

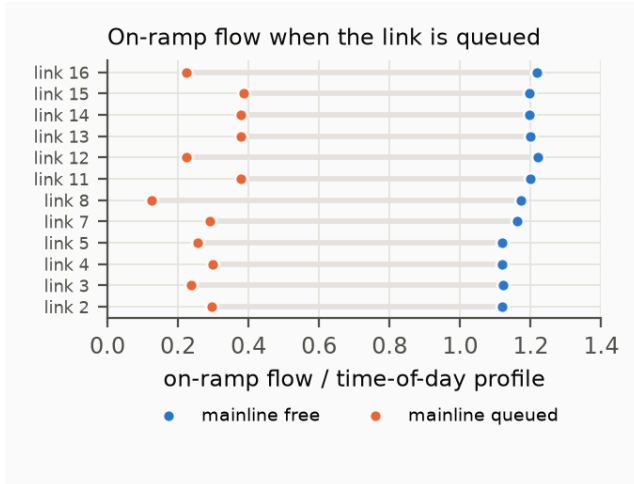


Figure 3. On-ramp flow relative to its time-of-day profile while the attached mainline link is free vs queued (D7_I405_S, train). The ramp layer stays published inside the Task 2 blackouts.

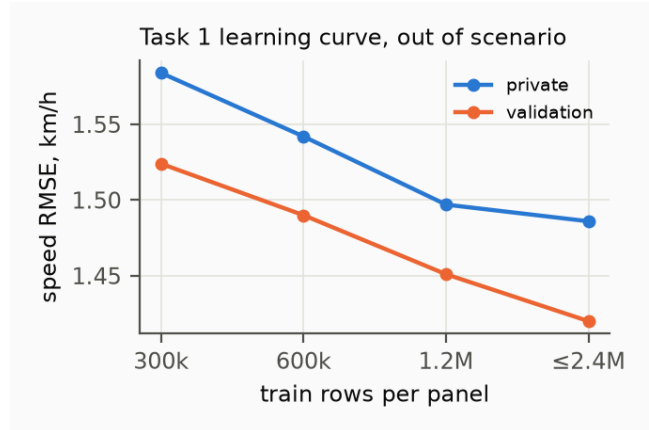


Figure 4. Task 1 speed RMSE out of scenario (hidden cells of validation/private) against the number of training rows per panel.

4.2 Results (out of scenario, hidden cells; RMSE, private / validation)

Main model	speed (km/h)	flow (veh/h/lane)
inherited model (train profiles, no common mode)	1.888 / 1.883	30.26 / 30.32
+ common mode, train plateau (V12)	1.914 / 1.892	30.26 / 30.32
+ common mode, split-own plateau (V12b)	1.584 / 1.524	30.26 / 30.32
600k rows/panel, 5000 rounds	1.542 / 1.490	30.03 / 30.03
1.2M rows/panel	1.497 / 1.451	29.82 / 29.82
all train targets (V13)	1.486 / 1.420	29.69 / 29.68

On true free-flow cells the final model reaches 1.145 km/h out of scenario. The ideal estimator "plateau + common mode" reaches 1.152, so free flow is at the noise floor. The remaining speed error comes from queue-edge cells: cells that are queued in time but not in space have RMSE 5-9.

Gap specialist (synthetic blackouts)	speed	flow
without ramps	6.32 / 6.50	68.3 / 68.9
ramps, train ramp profile (V12)	9.78 / 9.83	114 / 110
ramps, split-own ramp + speed/flow profiles (V12b, V13)	4.92 / 5.10	65.7 / 65.3

5. Task 3 — physical consistency

Task 3 is computed from the Task 1 answer: $k = q/v$ and $N = kL$, and the conservation residual is taken against organizer fluxes projected onto the observations. At a target cell with observed neighbours, the residual is about $2 \cdot |N \text{ error}|$. N error is dominated by flow noise, about 2% of q , against about 1.2% for speed. A flux-based estimate cannot help either: the 5-minute flux term $dt \cdot q$ is about 20 times N , so flow noise in the fluxes swamps it. The physics score therefore follows Task 1 accuracy, mostly flow. The flow gains of the capacity steps add roughly +0.0006 through S_{LWR} .

6. Task 2 — queue forecasting

6.1 Onset

In every onset window the queue first appears at the last horizon step ($T+30$), on a cluster of bottleneck links. The model has three parts:

- a cluster model for which cluster activates;
- a link model inside an active cluster;
- an unconditional link model.

Decoding draws joint Monte-Carlo samples and picks the set that maximizes expected IoU. The window history ends at $T-5$. The origin-row-free refit gained +0.034 online.

6.2 Ongoing

The ongoing forecast is $0.5 \cdot \text{LightGBM} + 0.5 \cdot (\text{mean of CNN seeds})$ per cell and horizon step, with threshold 0.5. The CNN is a 1-D dilated residual network over links, taking 13 history steps $\times$ channels. The LightGBM uses per-cell features of the queue geometry: distance to the head and tail, run lengths, the queue's movement over the last 6 slots, up/downstream flows, and same-day pre-origin episode features. **Origin row:** the masked layer publishes row T at about 58% coverage, with only Task 1 targets blank. Training and inference both use it. Paired evaluation of the LightGBM part: in scenario $+0.0102 \pm 0.0008$, validation $+0.0231 \pm 0.0025$, private $+0.0211 \pm 0.0026$, 7/8 panels.

Ongoing IoU, out of scenario (private / validation)	
inherited ongoing (no origin row)	0.8813 / 0.8772
LightGBM with origin row	0.8863 / 0.8832
CNN with origin row (2 seeds)	0.8822 / 0.8876
blend (final)	0.8986 / 0.8980

A third CNN seed adds $+0.0002 \pm 0.0006$, so the blend is saturated.

6.3 What limits onset

When the predicted cluster is right, onset IoU is 0.94–0.98; when it is wrong, 0.14–0.16. The wrong-cluster rate is 12% in the training scenario and about 20% in the new ones. Onset IoU is 0.893 in scenario and 0.765 out of scenario. We tested every causal source of scenario evidence; none passed the adoption rule (Section 8). Suppose every other component sat at a generous ceiling simultaneously:

- S_{state} 0.954, all cells at the white-noise floor;
- S_{physics} 0.77;
- S_{ODME} 1.0;
- ongoing IoU 0.91.

Then the public prize zone (0.917) would still need onset IoU ≥ 0.872 in an unseen scenario.

7. Task 4 — OD and path-flow estimation

The link counts are noise-free. We take the L2 projection of the split's own weak prior b onto $\{A f = c, f \geq 0\}$, solved through its dual. It equals the organizers' ridge reference (NNLS, $\lambda = 0.05$) to within 0.04% L1. The relative L2 deviation from the prior reproduces five online points: L2, KL, a rescaled prior, a per-path shift and a pooled prior. L2 is optimal under it. Every alternative was worse online, by 0.11 to 0.30 S_ODME.

8. What did not work (out of scenario unless stated)

- **Onset physics precursors** (flat diagram, flow/capacity): none exist.
- **Onset with ramp demand:** gained on mined events from one panel, then lost 0.009 online. This was small-sample luck.
- **Onset with prior-day evidence:** cluster day-rates, activation timing with 0–4 earlier days, and same-day early evidence. None was significant, and the splits disagreed.
- **Onset calibration** (temperature, shrinkage): validation and private disagreed.
- **Ramp features for the main Task 1 model:** mixed, 12/20 panels.
- **Queued-speed plateau feature:** worse, 6/20 panels.
- **Spatiotemporal CNN imputer for Task 1** (36-slot $\times$ corridor windows, factorised dilated convolutions): it fits the training scenario's spatial patterns and does not transfer. Out of scenario its flow RMSE is about $2\times$ LightGBM's, and every blend weight was worse. The ongoing CNN with early-day channels failed the same way.
- **More ongoing CNN seeds:** saturated. Threshold 0.45: $+0.0004 \pm 0.0003$ in scenario; not adopted.

9. Submission history (public leaderboard)

Version	Change	Public
v1	interpolation, persistence, L2	0.70508
v2	static onset set	0.81819
v3-v6	LightGBM T1/T2, gap specialist, retuning, full-train T1	0.85566 $\rightarrow$ 0.86864
v9b	onset cluster mixture, LightGBM + 4-seed CNN ongoing	0.87104
V10a	origin-row-free onset refit	0.87446
V12	common mode + ramp gap (train statistics) + origin row	0.87324
V12b	split-own plateaus and profiles	0.87900
V13	Task 1 on all train targets	0.88107

10. Reproducibility

trafficflow/reproduce.sh is the single entry point. It rebuilds V13 from the Kaggle release and the official toolkit (CPU only: 4 cores, 15 GB RAM, about 8–9 hours). Every step caches its result and resumes after interruption. Versions are in trafficflow/requirements.txt. Code is under Apache-2.0. No external data and no pretrained models.